

Introduction

Inflation has been one of the most important topics in the financial sector for a long time. Recently with the COVID-19 pandemic, we have seen surges in the price level larger than what we have seen in the past 60 years in the Netherlands. This study focused on forecasting inflation in the Netherlands using multiple more traditional methods and comparing them to Machine Learning (ML) models. We also look at the driving factors behind the inflation forecasts in the Netherlands according to the ML models.

- A dataset comprising 431 observations and 127 features divided into a training set, a less volatile test set and a more volatile test set.
- A comparison between benchmark models, such as Random Walk (RW), Autoregressive (AR) and Seasonal Autoregressive Integrated Moving Average (SARIMA), to ML models, such as Random Forrest (RF), LASSO, Support Vector Machines (SVM) and hybrid ML models.
- Finding the driving factors behind the inflation forecasts of the best performing ML models.

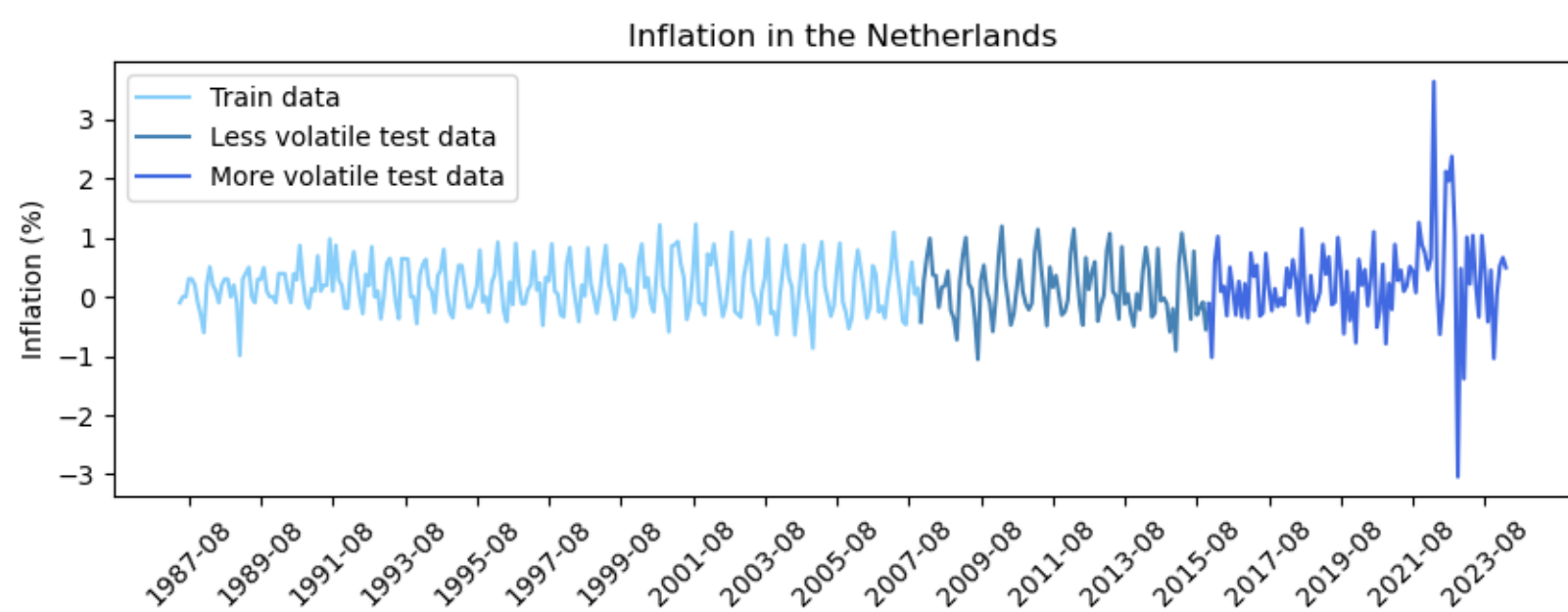


Figure 1. Monthly inflation in the Netherlands divided into three periods

Methodology

When forecasting inflation in a time-series setting we consider the following model based on $\mathbf{X}_t = (x_{1t}, \dots, x_{Mt})$ as an $M \times 1$ input vector which is mapped by the function $f(\cdot)$ which gives

$$\pi_{t+h} = f(\mathbf{X}_t) + u_{t+h}$$

where π_{t+h} is the inflation in month $t + h$ for $h = 1, \dots, 12$ and u_{t+h} is an error term. This gives the following forecasting equation

$$\hat{\pi}_{t+h} = \hat{f}(\mathbf{X}_t)$$

where $\hat{\pi}_{t+h}$ is the forecasted inflation for period $t + h$ and $\hat{f}$ is the estimated function based on data from $t - R$, with R being the window size. The function depends on the model used.

- Benchmark models** are the more traditional linear models.
 - Random Walk** is the most naive solution and predicts the last observed value.
 - Autoregressive** is a model based only on the lags of the inflation.
 - SARIMA** is an extension of the AR model which can capture seasonal trends and more.
- ML models** are often non-linear models which apply different methods.
 - RF** uses multiple regression trees and takes the average prediction over the trees.
 - XGBoost** is a modified version of RF which uses approximations to speed up computations.
 - LASSO** is a linear shrinkage method which shrinks the irrelevant features to zero.
 - SVM** finds a hyperplane which maximises the margin between the subsets of data and the hyperplane.
 - Hybrid models** first use a shrinkage method to select the relevant features and then use a ML model to forecast inflation, here LASSO-XGBoost is used.

Shapley values are used to find the driving factors of the inflation. A prediction for inflation can be decomposed into the sum of Shapley values. The Shapley value of feature m is based on the difference in prediction power between a model containing feature m and a model without it. Using these Shapley values we can compute the Weighted Correlation Coefficient (WCC), which averages the Shapley values and scales them to gain a sense of average impact for each feature.

Results: Forecasting

The results regarding the forecasting power of the models are summarized in Table 1. The results are based on the Root Mean Squared Error (RMSE), the Mean Absolute Error (MAE) and the Median Absolute Deviation from the median (MAD). The results are scaled to the results of the RW model.

Model	Forecasting precision					
	Less Volatile			More Volatile		
	ave. RMSE	ave. MAE	ave. MAD	ave. RMSE	ave. MAE	ave. MAD
RW	1.000	1.000	1.000	1.000	1.000	1.000
AR (3)	0.731	0.737	0.727	0.754	0.733	0.720
AR (12)	0.478	0.452	0.433	0.810	0.694	0.504
SARIMA	0.470	0.452	0.432	0.734	0.642	0.469
RF	0.606	0.600	0.599	0.738	0.669	0.597
LASSO	0.615	0.617	0.642	0.761	0.716	0.645
SVM	0.629	0.620	0.604	0.756	0.730	0.688
LASSO-XGBoost	0.611	0.602	0.591	0.726	0.678	0.651

Table 1. Average RMSE, MAE and MAD based on all forecasting horizons for all models for the less volatile and more volatile test sets.

Results: Forecasting

- In general the models perform worse in more volatile times.
- All ML models are outperformed by the SARIMA model and often also the AR(12) model.
- The regression tree-based models seem to perform best among the ML models.
- The use of a feature selection technique before estimating does not yield significant improvements.

In comparison to other literature with similar studies, the results of forecasting have improved [1, 2]. Some potential reasons for this could be:

- The inclusion of additional lags of the inflation which are added to the features.
- The inflation in the Netherlands is easier to model than in other countries. This can be seen by the fact that the AR(3) model performs better here than in other literature.
- Seasonality in the data which is recognized and used to improve the results.

Results: Explaining

The results of the explanation of the inflation are summarized in Figure 2. This shows the change in feature importance over time according to the WCC, based on the Shapley values. The models selected for each forecasting horizon were the best-performing ML models for those horizons respectively.

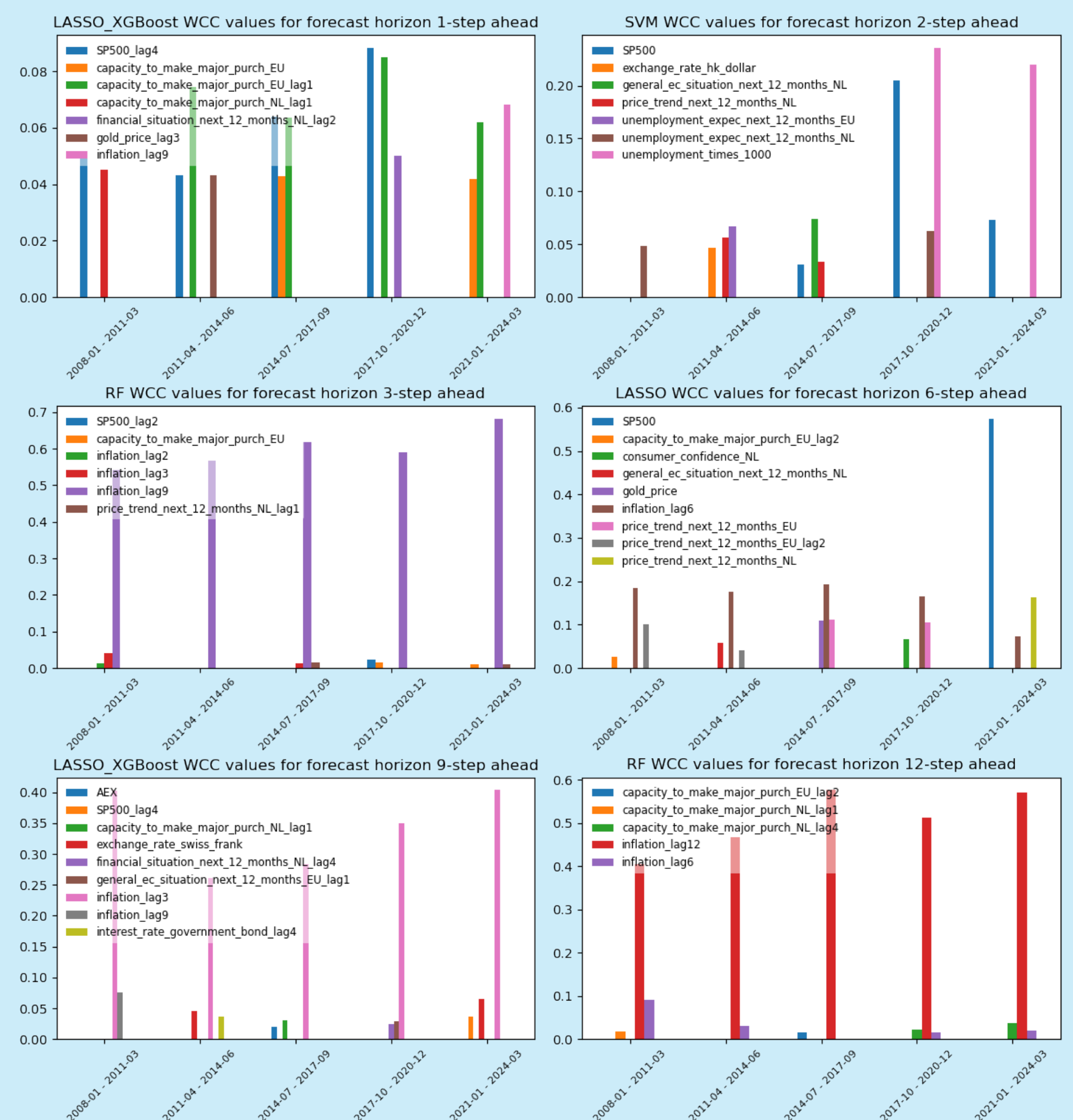


Figure 2. The WCC showing the change in feature importance over time for the best model for each forecasting horizon

- The lagged inflation is very important for forecasting inflation.
 - The inflation from 12 months ago is the most important for forecasting, this can be clear from the 3, 6, 9, and 12-step ahead forecasts.
 - There is a positive relationship between the current inflation and the inflation from 6 and 12 months ago, while there is a negative relation with the inflation 3 and 9 months ago.
- The regression tree-based models rely heavily on lagged inflation in comparison to SVM and LASSO.
- Expectations people have for the future regarding the economy and their capacity to adapt to it are other driving factors of the ML models.

Conclusion

Forecasting inflation in the Netherlands is a task that ML models are capable of doing and can be used to gain more insight into the driving factors of inflation.

- ML models are capable of forecasting inflation, but in case of seasonality, they will get outperformed by models catered to this. However, the performance of ML models is more robust to volatile times.
- There is seasonality present in the inflation in the Netherlands, with a cycle every 12 months.
- The main driving factors of inflation are the previous inflation, the expectations people have regarding the economy and their ability to adapt to this.

References

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- Marcelo C. Medeiros, Gabriel F. R. Vasconcelos, Álvaro Veiga, and Eduardo Zilberman. Forecasting inflation in a data-rich environment: The benefits of machine learning methods. *Journal of Business & Economic Statistics*, 39(1):98–119, 2021.